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The Humanitarian Sector by the Numbers: A Business Report on the GDHO 2025 Dataset

A plain-language summary for donors, analysts, and humanitarian-sector decision-makers. Technical detail (methodology, statistical tests, model diagnostics) lives in `outputs/insights.md` *and* `outputs/ml_results.md`*; this report translates those findings into what they mean and what to do about them.*

What this data is, and why it matters

The Global Database of Humanitarian Organisations (GDHO) tracks close to every organization that does humanitarian work worldwide — national NGOs, international NGOs, Red Cross/Crescent societies, and UN agencies — from 2006 through 2024. For each organization and year, it records staff numbers, budget, sector of work, headquarters, and every country the organization operates in.

For anyone who funds, coordinates, or studies humanitarian response — donors, UN coordination bodies, INGOs choosing local partners, researchers — this is one of the only datasets that lets you see the *whole* sector at once, not just the large, well-known organizations that publish their own financial reports. That is also exactly its limitation: because most organizations in it don't publish detailed financials, GDHO has to estimate a great deal of what it reports. This report treats that limitation as a first-class finding, not a footnote.

The shape of the sector

It's a long tail, not a level playing field. In 2024, 81% of tracked organizations were national NGOs (NNGOs) — locally rooted, usually small. But those same organizations accounted for only 3.4% of total reported operating budget. Typical budget size varies by roughly 6,780x between a median national NGO (~$528,000/year) and a median UN agency (multi-billion dollar). Nearly all humanitarian *capacity*, measured in dollars, sits with a small number of very large actors, even though most *organizations* doing the work are small and local.

The sector may have peaked in 2020. The number of tracked organizations grew every year from 2006 to 2020 (6,225 → 7,165), then fell 21% to 5,662 by 2024. That decline lines up with the pandemic and its aftermath, but the data can't say for certain how much is real sector contraction versus organizations simply not reporting in recent years — that ambiguity itself is worth flagging to anyone using the recent years' totals as a measure of sector health.

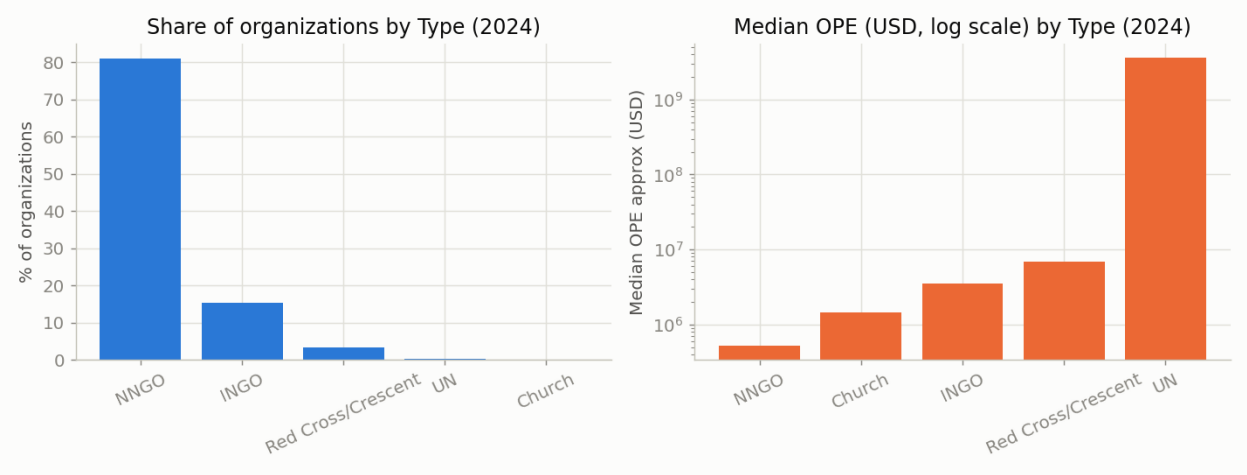


Figure 1: Share of organizations by Type, and median budget by Type

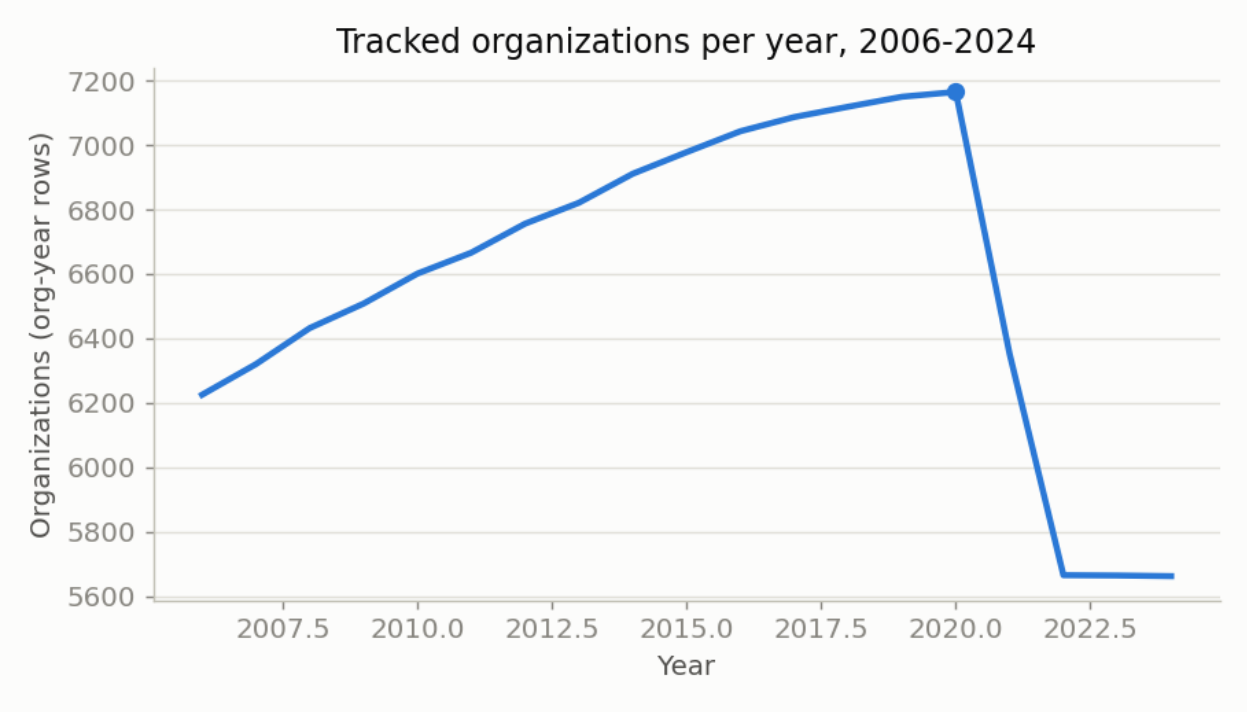


Figure 2: Tracked organizations per year, 2006-2024

Where the work happens is not where the money comes from. The organizations most frequently headquartered somewhere are headquartered in Somalia, the United States, Pakistan, Afghanistan, and South Sudan — crisis-affected countries themselves, not traditional donor capitals. That's a direct consequence of NNGO dominance: most organizations are headquartered exactly where they work. Operational presence concentrates in a familiar list of protracted crises — Afghanistan, Pakistan, Somalia, DR Congo, Haiti, Kenya.

Top 5 HQ locations (2024)	Orgs	Top 5 operation countries (2024)	Orgs
Somalia	500	Afghanistan	350
United States	334	Pakistan	303
Pakistan	260	Somalia	268
Afghanistan	254	DR Congo	244
South Sudan	191	Haiti	239

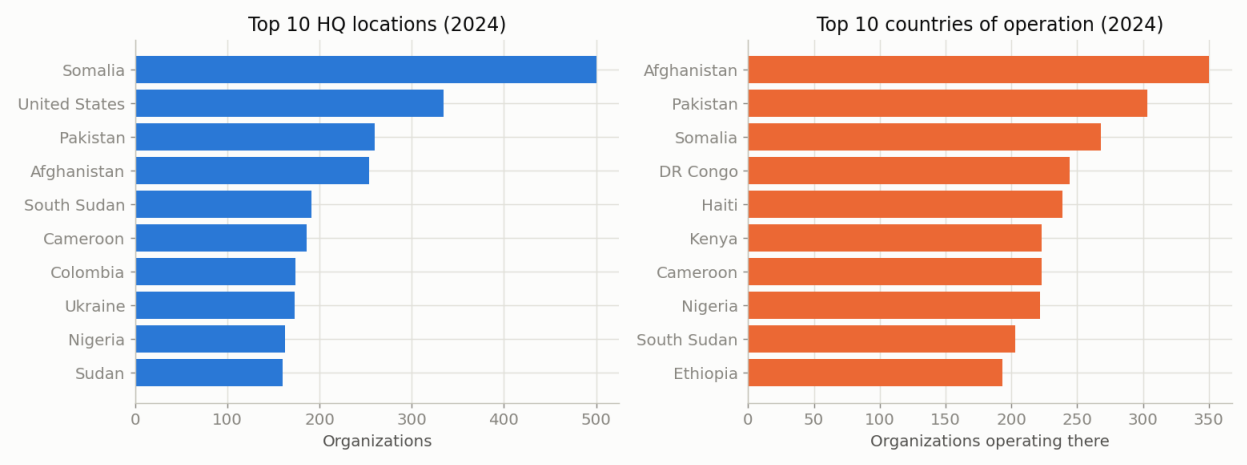


Figure 3: Top 10 HQ locations and top 10 countries of operation

Budget size doesn't tell you how "humanitarian" an organization is. Organizations with very similar total budgets can spend anywhere from about half to two-thirds of it on humanitarian work specifically (the rest going to development, advocacy, or other non-humanitarian programming) — and in some cases the gap is far wider. Budget alone is not a reliable proxy for humanitarian focus; anyone screening organizations for humanitarian-specific funding should look at the humanitarian-share field directly, not total spend.

The data-quality picture: read the numbers with appropriate caution

This is the part of the analysis most relevant to *how* this data should be used, not just what it shows:

- **Only about 7% of budget figures and 5% of staff figures are directly reported.** Everything else — the overwhelming majority — is GDHO's own estimate. Totals and averages drawn from this dataset are best understood as **modeled estimates of the sector**, not an audited census.

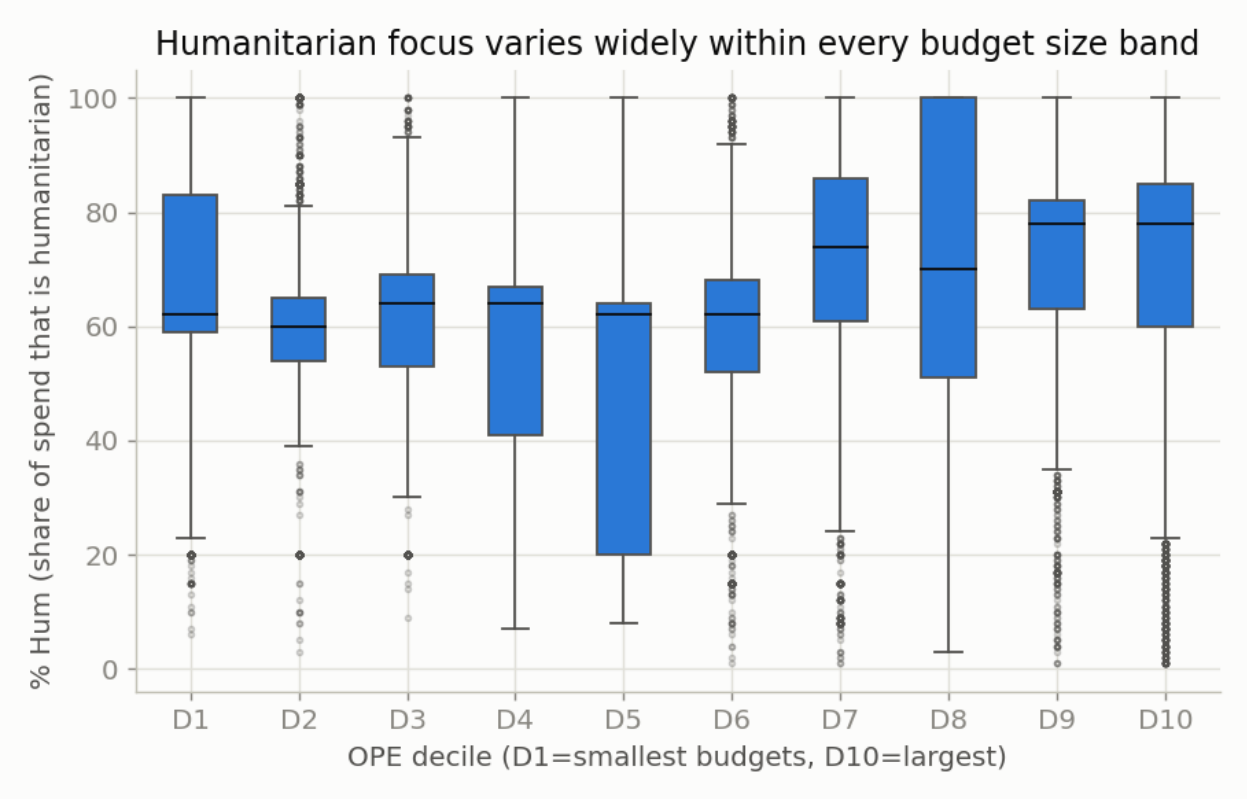


Figure 4: Humanitarian focus varies widely within every budget size band

- **Hundreds of organizations share an identical budget figure.** In 2024 alone, one specific dollar figure was assigned to 882 different organizations. This is strong evidence that GDHO's estimation approach assigns a shared, bucketed figure (by organization type, budget tier, region, and year) to organizations that don't report their own numbers, rather than estimating each one individually. It's a reasonable approach for filling a gap at scale, but it means budget figures for any *specific* small/unreported organization should be treated as a bucket average, not that organization's real number.

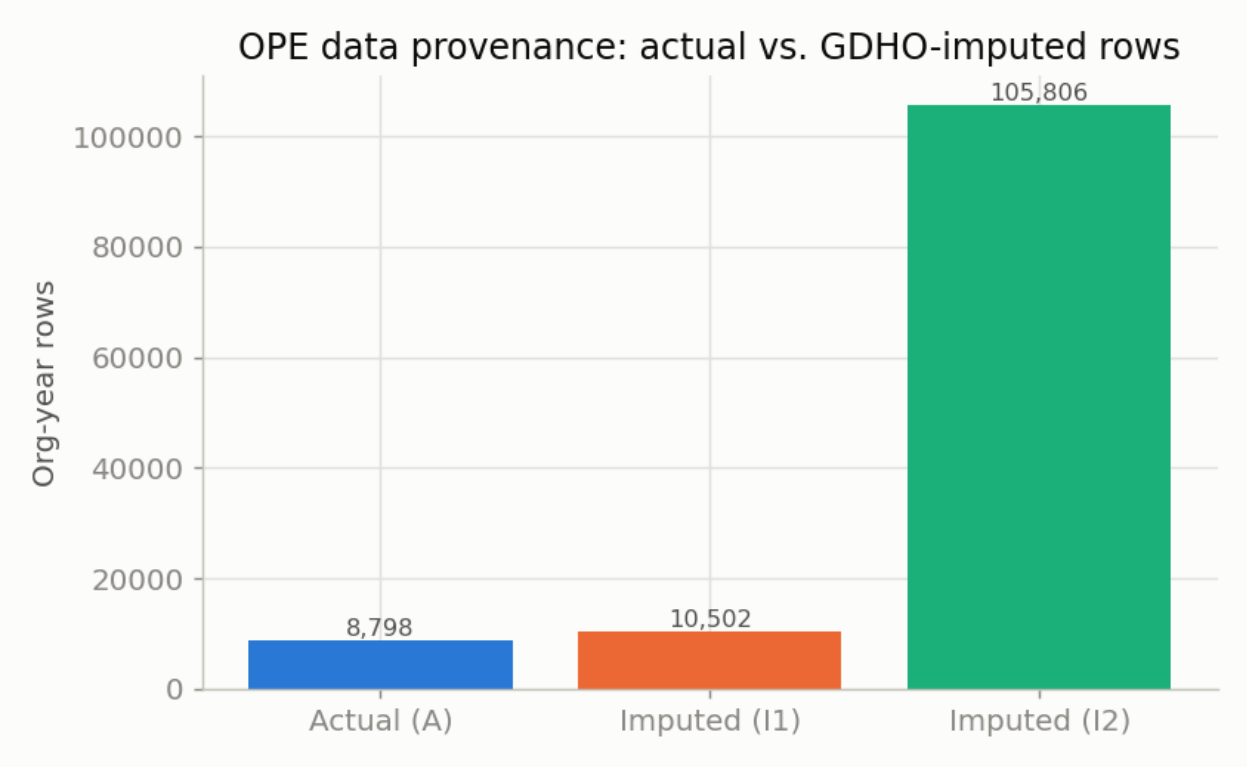


Figure 5: OPE data provenance: actual vs. GDHO-imputed rows

- **A key classification field is missing for over a quarter of records.** Whether an organization is "International" or "National" is blank 26.5% of the time — but it's almost perfectly predictable from the organization's Type (INGO vs. NNGO), so this is a cheap, low-risk data-quality fix available to anyone maintaining this dataset going forward.
- **Sector and religious-affiliation fields are sparse** (recorded for roughly 8% and 13% of organizations respectively) — treat any sector- or affiliation-based breakdown as directional, not comprehensive, since it only reflects the subset that happened to report it.
- **A handful of fields need care when compared across years** — budget figures should be compared in inflation-adjusted terms (nominal figures understate older years by up to ~56% relative to 2024 dollars), and the "countries of operation" count field doesn't always match the underlying country checklist, so it should be treated as approximate.

None of this makes the dataset unreliable for its intended purpose — sector-wide, directional analysis — but it does mean it should not be cited as precise financial reporting for any single organization.

A practical tool: an independent budget estimator

Given how much of the dataset's own financial data is already estimated rather than reported, we built and tested a machine-learning model that estimates an organization's likely operating budget from its *operational profile* — its type, staffing levels, region, sector, and how many countries it works in — without looking at any financial figure at all.

Why this is useful: anyone trying to size an organization they don't have financial data for (a donor vetting a new local partner, a coordination body mapping capacity in a region) can get a reasonable estimate from operational characteristics alone. It also serves as an independent cross-check against GDHO's own estimates.

How well it works: on organizations the model had never seen before (the honest test — see `outputs/ml_results.md` for why this matters), the best model explained about 80% of the variation in budget size, with a typical error in the mid-single-digit percent range on the "typical" organization (errors are naturally much larger for a handful of extreme outliers — a multi-billion-dollar UN agency is harder to pin down precisely than a $500,000 local NGO).

The honest caveat: part of that accuracy comes from the model learning to recognize GDHO's own bucketed estimation approach (see the data-quality point above), not from discovering some independent truth about each organization's finances. That's not a flaw in the model — it's exactly what makes it useful as a fallback estimator — but it should not be marketed as more than that.

Model	Held-out test R^2	Test MdAPE
LightGBM (best)	**0.796**	**2.8%**
Random Forest	0.790	0.0%
Ridge (linear)	0.543	35.5%
Dummy (median baseline)	-0.049	66.7%

Staff headcount is, by far, the single strongest predictor of budget size — followed by region and the year of observation. This is intuitive (bigger operations need more people) but useful to know explicitly: if a donor or coordination body can collect just one additional data point about an unreported organization, staff count is the highest-value one to prioritize.

Recommendations

1. **Report sector totals as modeled estimates, with the ~7%-actual caveat attached**, whenever this data is cited in donor communications, situation reports, or sector-mapping products.
2. **Backfill the "International or national" field using organization Type** — a low-effort, high-confidence fix that would resolve the classification for most of the 26.5% currently missing.
3. **Prioritize collecting Sector and staff-count data** for unreported organizations over collecting budget figures directly — sector coverage is currently thin (~8%), and staff count is both easier to obtain and the strongest available signal for budget size.
4. **Use the budget estimator (`outputs/budget_estimator_model.joblib`) as a screening tool, not a final figure** — it's well suited to flagging "this organization looks larger/smaller than its peers" or filling a gap for sector-wide totals, not for citing a specific dollar amount for a specific organization in a funding decision.

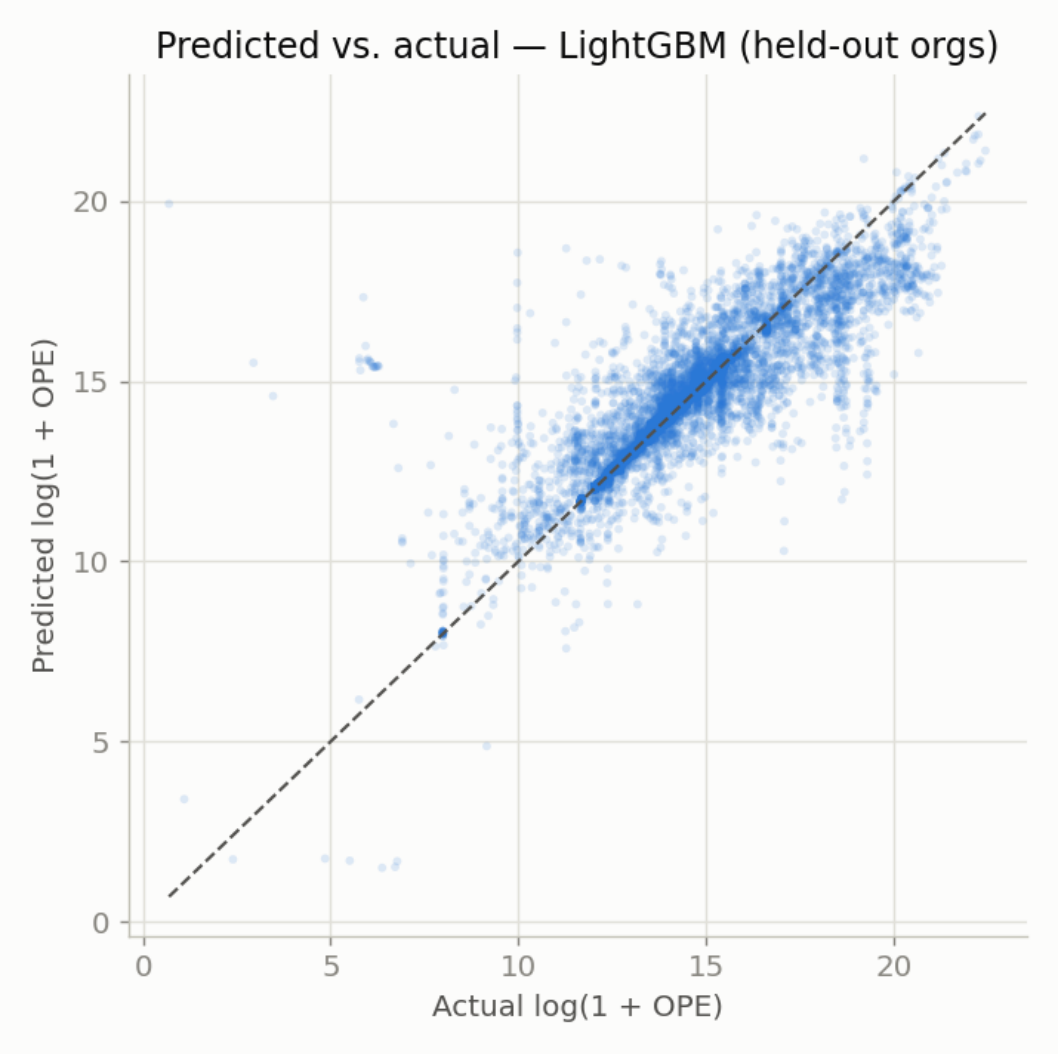


Figure 6: Predicted vs. actual budget, held-out organizations

5. **Always compare budgets across years in inflation-adjusted terms**, and treat the "countries of operation" count as approximate rather than exact, given the discrepancies documented above.
6. **Investigate the post-2020 decline in tracked organizations** before treating it as evidence of sector contraction — distinguishing real attrition from reporting gaps would materially change how this trend should be communicated.

Where to go for more detail

- `outputs/insights.md` — all 20 data insights, each with the exact figures and a supporting chart.
- `outputs/ml_results.md` — full modeling methodology, metrics, and limitations for the budget estimator.
- `README.md` — how this analysis was built and how to reproduce it.